

VESPA portal: a single-entry point to all validated EPN-TAP data services

Locate individual data products meeting certain conditions (observational or instrumental)

Typical request for surfaces:

Mars, a given region (Tharsis volcanoes area) — images, spectra or spectral-cubes

Illumination conditions ($i \leq 20^\circ$, phase $\leq 50^\circ$)

Local time or season range (L_s)

Minimum resolution

Spectral range / filter

an instrument

etc

The portal displays a simple parameter form to build the query

Returns results from each service
=> described uniformly

Resulting
TAP query

All results in a single table

<https://vespa.obspm.fr>

The screenshot shows the VESPA portal interface. At the top, there's a header with the VESPA logo and the text "Virtual European Solar and Planetary Access". Below the header, there's a "Refine your search" section with various input fields for parameters like Target Name, Target Class, Dataproduct Type, Instrument Host Name, Instrument Name, Processing level, Time, Location, Spectral, Illumination, Data Reference, and Optional. To the right of the search parameters, there's a "Data Services" section listing various data services with their respective result counts. Below the data services, there's a "Generated WHERE clause of ADQL statement:" section showing the resulting TAP query. At the bottom, there's a "Reset" button and a "Submit" button.

Annotations on the screenshot:

- "Examine service results" points to the "Data Services" section.
- "Save on disk" points to the download icon in the top right corner.
- "Send to VO apps" points to the "Send to VO apps" button in the top right corner.
- "Resulting TAP query" points to the "Generated WHERE clause of ADQL statement:" section.
- "All results in a single table" points to the "EPN-TAP compilation results" section at the bottom.

Data Service	Results
abs_cs - Data for numerical modeling of planetary atmospheres	13 results
APIS - Auroral Planetary Imaging and Spectroscopy	74175 results
BaseCom - The Nançay Cometary Database	6886 results
bass2000 - Bass2000 solar survey archive	357033 results
BDIP - IAU database of historical planetary images	16906 results
cassini_jupiter - Cassini RPWS/HFR Calibrated Jupiter Flyby Dataset	7 results
CLIMSO - CLIMSO coronagraphs at Pic du Midi de Bigorre	1021345 results
cpstasm - CLUSTER STAFF-SA Spectral Matrix Data	11688 results
DynAstVO - Asteroid orbital database and ephemerides	29733 results
eit_syn - Synchronous synoptic maps of the solar corona from EIT/SoHO	18482 results
ExoPlanet - Extrasolar Planets Encyclopaedia	5177 results

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SELECT * FROM ... WHERE ((c1min <= c1max AND c1min <= 265.0 AND c1max >= 235.0) OR (c1min > c1max AND (c1min <= 265.0 OR c1max >= 235.0))) AND c2max >= -15.0 AND c2min <= 15.0 AND (1 = ivo_hashlist_has(lower("target_name"),lower('Mars')) OR 1 = ivo_hashlist_has(lower("target_name"),lower('4')) OR 1 = ivo_hashlist_has(lower("target_name"),lower('499')) AND "incidence_min" <= 20.0 AND "spatial_frame_type" = 'body'
```

EPN-TAP compilation results